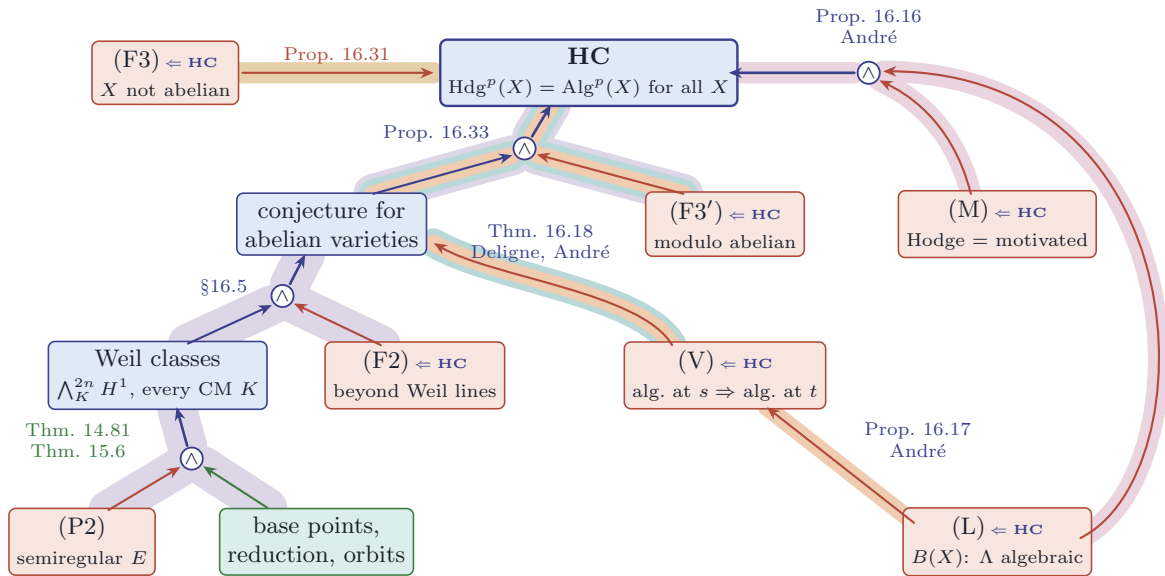




the five minimal sufficient sets of the closure theorem, drawn as nested ribbons above; routes 3 to 5 end with $\text{ab.} \wedge (\text{F3}') \Rightarrow \mathbf{HC}$

open proved here derived $\Leftarrow \mathbf{HC}$ implied by $\mathbf{HC}$ $\bigwedge$ rule with several premises